

Customer Behavior Analysis

InternSpark Data Analytics Internship – Task 1

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Task: Customer Behavior Analysis

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Dataset Source: Kaggle – Customer Behavior Analysis Dataset

1. Executive Summary

This project focuses on analyzing customer purchasing behavior using an e-commerce dataset. The analysis involved data cleaning, feature engineering, customer segmentation using RFM (Recency, Frequency, Monetary) analysis, visualization of purchasing patterns, and churn analysis. The insights obtained from the analysis can help Alfido Tech improve customer engagement, increase customer retention, and make data-driven business decisions.

2. Objective

The objective of this project is to analyze customer transaction data to identify customer segments, purchasing patterns, and churn risks. The project aims to provide actionable recommendations that can help improve customer engagement and business performance.

3. Dataset Overview

Dataset Name: Customer Behavior Analysis

Source: Kaggle

Description:

The dataset contains customer transaction records including purchase dates, product categories, payment methods, customer demographics, purchase amounts, and churn information.

Key Columns Used

- Customer ID
- Purchase Date
- Product Category
- Product Price
- Quantity
- Total Purchase Amount
- Payment Method

- Customer Age
- Gender
- Returns
- Churn

4. Data Cleaning

The following preprocessing steps were performed:

- Checked the dataset structure and data types.
- Removed duplicate records.
- Converted the Purchase Date column into datetime format.
- Checked for missing values.
- Handled missing values in the Returns column.
- Removed unnecessary columns (if applicable).

5. Feature Engineering

New features were created to improve analysis.

- Extracted Year from Purchase Date.
- Extracted Month from Purchase Date.
- Extracted Day from Purchase Date.
- Created Average Purchase per Item.

These features helped analyze purchasing behavior over time.

6. Customer Segmentation (RFM Analysis)

Customers were segmented using the RFM model.

Recency

Measures how recently a customer made a purchase.

Frequency

Measures how often a customer makes purchases.

Monetary

Measures the total amount spent by a customer.

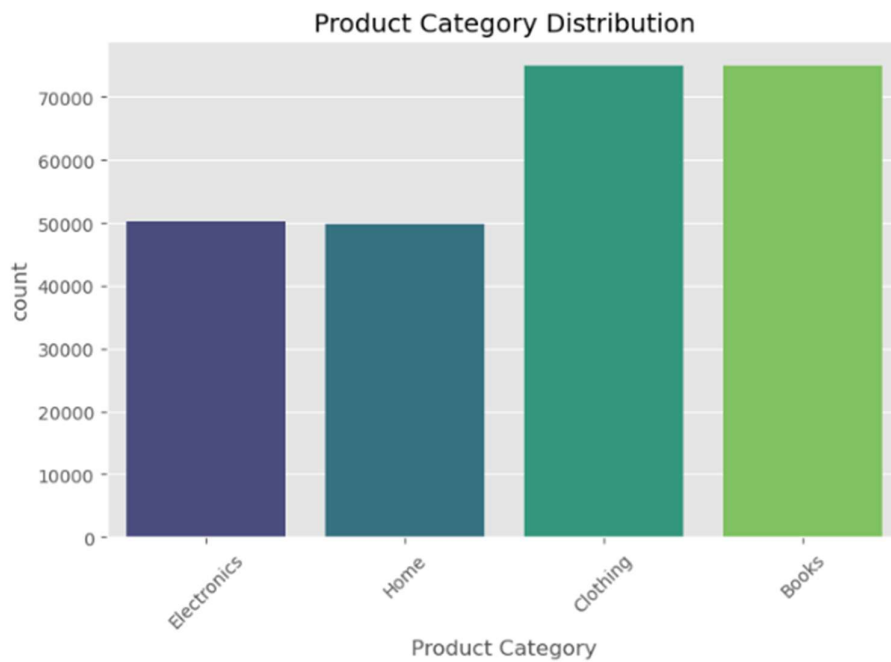
Based on these metrics, customers were classified into the following segments:

- Champions
- Loyal Customers
- Potential Loyalists
- At Risk Customers
- Lost Customers

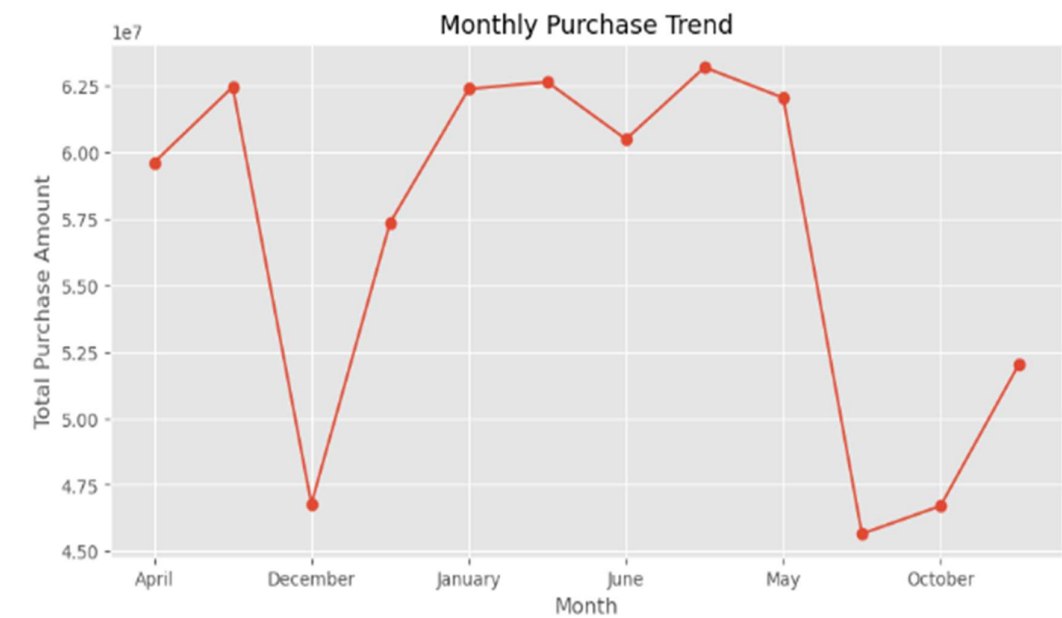
This segmentation helps businesses target customers with personalized marketing strategies.

7. Visualizations

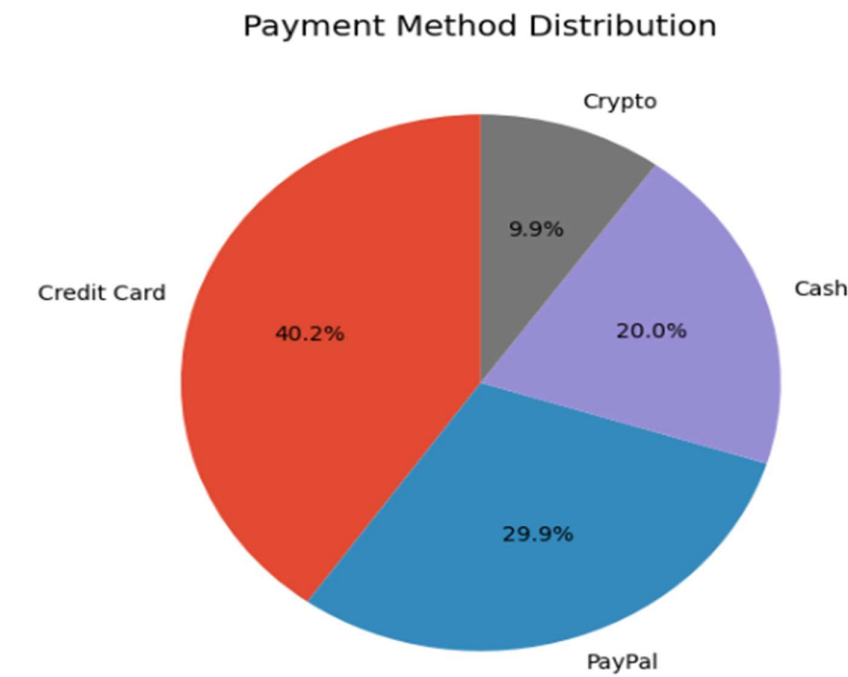
Product Category Distribution



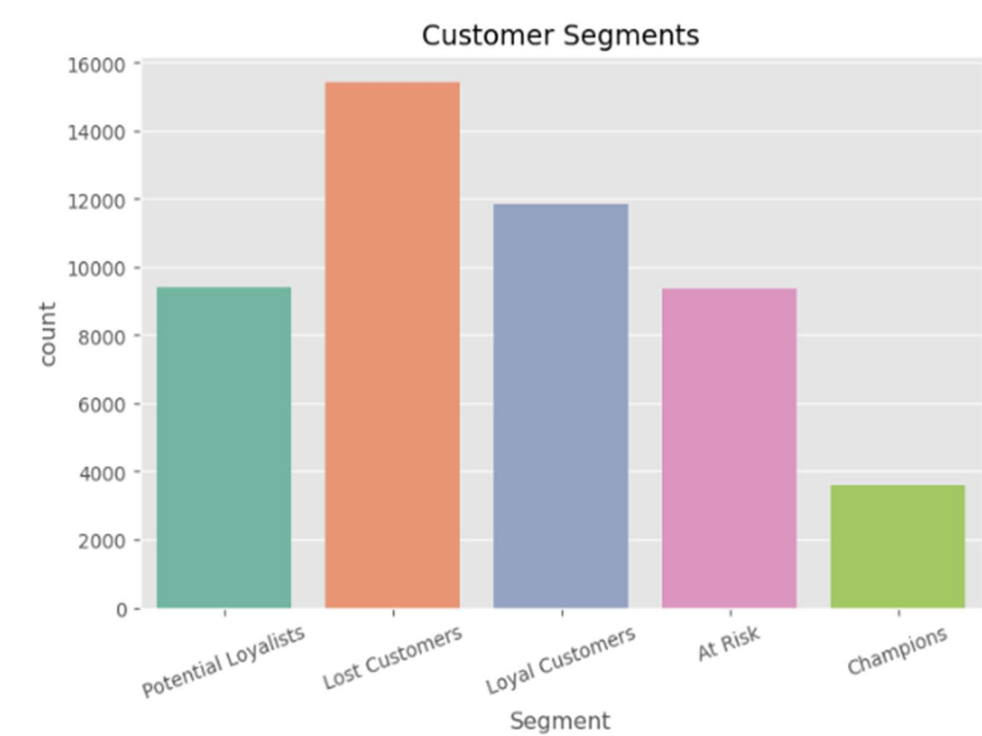
Monthly Purchase Trend



Payment Method Distribution



Customer Segment Distribution



Revenue by Product Category



8. Key Findings

- The dataset contains customer transaction details such as purchase amount, product category, payment method, and churn status.
- RFM analysis segmented customers into Champions, Loyal Customers, Potential Loyalists, At Risk, and Lost Customers.
- Certain product categories generated higher revenue, indicating stronger customer demand.
- Customer churn analysis identified customers who may require targeted retention strategies.
- Monthly purchase trends revealed fluctuations that can help plan marketing campaigns and inventory management.

9. Recommendations for Alfido Tech

- Introduce a loyalty rewards program for Champions and Loyal Customers to encourage repeat purchases.
- Launch personalized email or SMS campaigns targeting At Risk and Lost Customers with exclusive offers.
- Promote high-performing product categories through targeted advertisements and attractive bundle deals.
- Analyze customers with high churn rates and implement retention strategies such as personalized recommendations and discounts.
- Optimize marketing campaigns based on seasonal purchase trends to maximize customer engagement and increase revenue.

10. Conclusion

- This project successfully analyzed customer purchasing behavior using data cleaning, feature engineering, RFM segmentation, and visualization techniques. The findings highlighted valuable customer segments, purchasing trends, and churn risks. The recommendations provided can help Alfido Tech improve customer satisfaction, strengthen retention strategies, and support better business decision-making through data analytics.